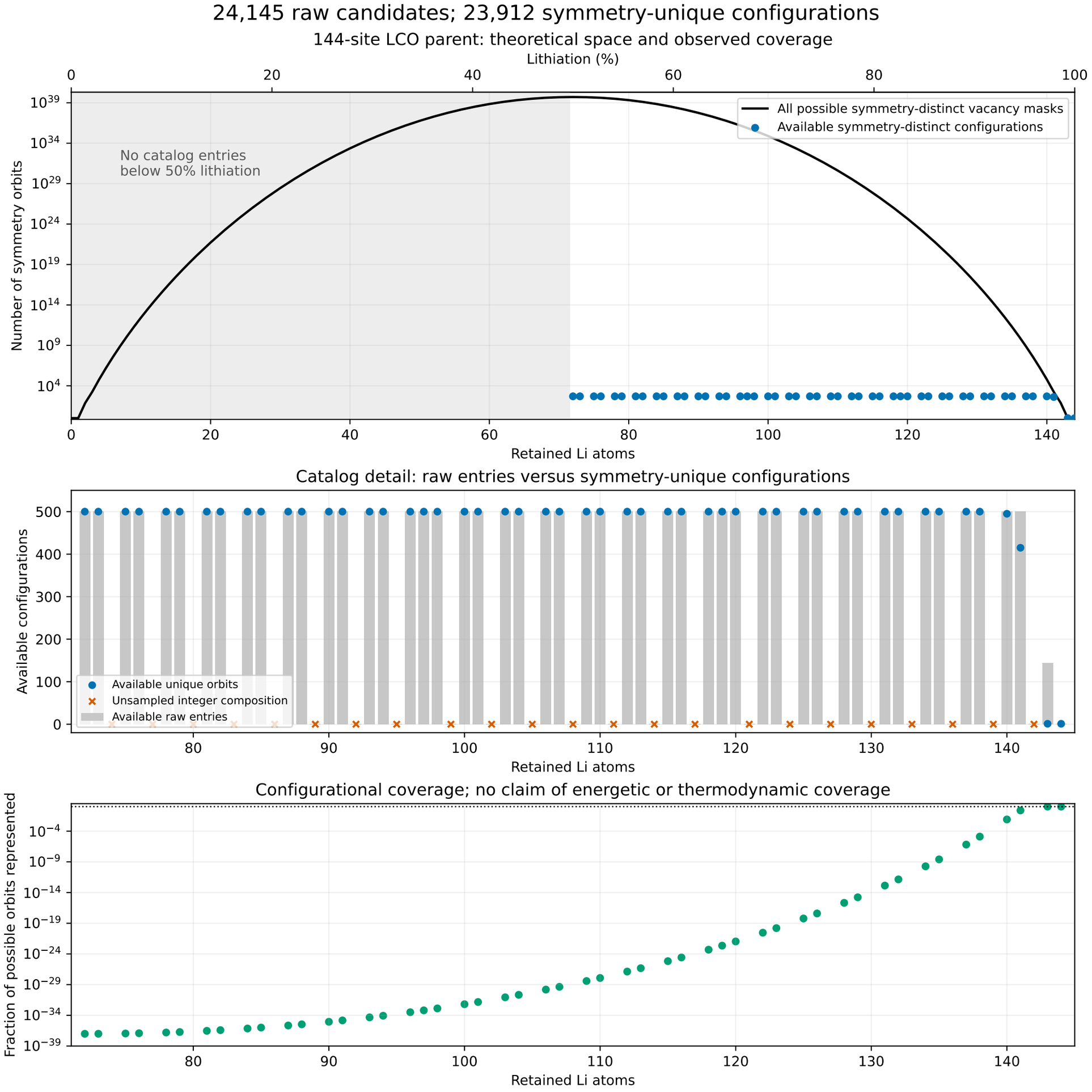
24,145 raw candidates; 23,912 symmetry-unique configurations
144-site LCO parent: theoretical space and observed coverage
Lithiation (%)
0
20
40
60
80
100
Number of symmetry orbits
$10^{39}$
$10^{34}$
$10^{29}$
$10^{24}$
$10^{19}$
$10^{14}$
$10^{9}$
$10^{4}$
No catalog entries
below 50% lithiation
All possible symmetry-distinct vacancy masks
Available symmetry-distinct configurations
0
20
40
60
80
100
120
140
Retained Li atoms
Catalog detail: raw entries versus symmetry-unique configurations
Available configurations
500
400
300
200
100
0
Available unique orbits
Unsampled integer composition
Available raw entries
80
90
100
110
120
130
140
Retained Li atoms
Configurational coverage; no claim of energetic or thermodynamic coverage
Fraction of possible orbits represented
$10^{-4}$
$10^{-9}$
$10^{-14}$
$10^{-19}$
$10^{-24}$
$10^{-29}$
$10^{-34}$
$10^{-39}$
80
90
100
110
120
130
140
Retained Li atoms